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Study of the plasma pinch and ion beam properties versus the nitrogen gas pressure using the Lee model code

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Abstract:

This study presents the results of a series of numerical experiments that were carried out using Lee code in order to study the properties of plasma pinch versus nitrogen gas pressure and its effect on the number and energy of ion beams emitted from the two dense plasma focus devices PF400 and APF. The results showed that the number of nitrogen ions are 7.9×10^{13} ions (for PF400) and 3.82×10^{14} ions (for APF), while the highest beam energy is 18 J for APF, due to the effect of the operating energy and the parameters of the two studied devices on the properties of the ion source (the pinch). This study presents the properties of an ion source that can be used in practical applications.

Article Highlights:

The plasma pinch and ion beam properties versus nitrogen gas pressure.

Dense plasma focus devices as ion sources for various applications.

Keywords: The Lee model code, plasma pinch, ion beam, dense plasma focus.

1- Introduction:

Dense plasma focus is a type of plasma that was originally developed as a Z-pinch device for nuclear fusion in the early 1960s, The initial principles of the plasma focus phenomenon were laid by N.V. Filippov in 1950 in the Soviet Union, where he developed a design for a dense plasma focus device, and around the same period a different version of these devices was developed by J.W. Mather in the United States of America[1]. In the dense plasma focus device, there are two cylinders united in the center, forming the electrodes, where the inner cylinder is an anode and the outer cylinder is a cathode separated by an insulator as shown in the figure 1.

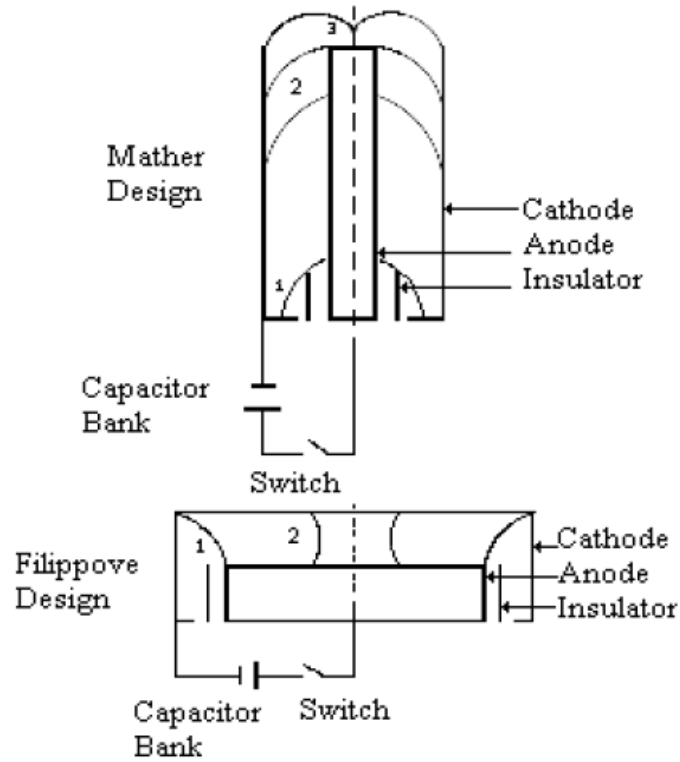


Fig 1: Scheme showing the geometry of the Mather model (up) and the Filippov model (down) [2]

These two cylinders are placed in a vacuum chamber, then the gas is pumped under a certain pressure, and then a high-intensity electric current (of the order of kiloampere) is passed through the discharge of a capacitors bank connected to the chamber, which leads to ionization of the gas and the current begins to move from the anode to the cathode, then the emergence of a magnetic field pushes the plasma from the bottom of the chamber to the top axially, and when it reaches the end of the electrodes, it continues to move slightly upward and then begins to bend inward in the form of an umbrella where the plasma is compressed and its density increases dramatically as a small column of plasma is formed or what is known as pinch, The speed of the plasma during these stages is several times the speed of sound. This column lasts for a very short period of time (on the order of nanoseconds) and then collapses due to the instability of the plasma, and its collapse generates intense electromagnetic radiation and beams of particles (ions and neutrons) [3]. The dense plasma focus was distinguished by the quality and density of the radiation emissions, which include X-rays and ions and electrons beams according to the type of operating gas used in addition to the characteristics of the device in terms of the engineering of the vacuum chamber and parameters of the electrical circuit. Therefore, a large number of experimental and simulation studies were conducted to study these emissions and the factors affecting them, in addition to studying the possibility of using them in different applications. In plasma focus X-ray studies Bogolyubov et al. [4] used soft X-rays from 15 KJ plasma focus device to electronic printing in the range of wavelengths (9-14 Å) where the highest value of yield 100 J and a discharge current of 280 kA. Zakaullah et al. [5] used an plasma focus device working with argon gas as a source for soft X-rays at different operating pressures, where the highest yield

was $Y_{\text{SXR}}=30$ mJ at 1.5 mbar efficiently 0.0015%. Akel et al. [6] studied by carrying out numerical experiments the dependence of the soft X-ray yield of the argon plasma on the energy stored in the capacitor bank, total current, and pinch current. Apilado et al. [7] studied the correlation of the neon SXR pulse from INTI device with plasma focus dynamics. Singh et al. [8] performed numerical experiments using Lee code to obtain the scaling laws for peak current, pinch current and energy for five Plasma focus devices. Lim et al. [9] conducted an experiment using a Mather-model plasma focus device that operates with deuterium gas as a source of deuterium ion beams at a pressure range (0.05-0.5 mbar), and it was concluded that the number of deuterium ions per shot is 10^{18} ions. Akel et al. [10] conducted numerical experiments using the modified Lee code on different plasma focus devices running on nitrogen gas to study the properties of the outgoing ion beam, where the study showed that the ion energy decreases with increasing gas pressure. While the number of ions increases with the increase in gas pressure and the highest value of energy density falls within the range (10^{12} - 10^{14} W.m⁻²). Saw et al. [11] established the measurement laws for the properties of the beam of deuterium ions from the plasma fist, where it was found that the number of deuterium ions falls within the range (12×10^{14} - 40×10^{14}) ions for each kJ and that the beam energy constitutes (1-5%) of the operating energy of the incinerator plasma device, and the energy density is within the range (3×10^{13} - 57×10^{13} W.m⁻²). Akel et al. [12] carried out numerical experiments using a Lee code to calculate the properties of the ion beams emitted by several plasma focus devices when using deuterium gas. Roshan et al. [13] studied the using of high-energy deuterons beams from a low-energy plasma focus device to activate a boron carbide target. Soltani et al. [14] also analyzed the angular distribution of deuterium ions from a 3.4 kJ dense incinerator plasma device and an ion was obtained in the shot. Also Aghamir et al. [15] performed a comparison of carbon formation on aluminum and graphite substrates using a low-power device. The characteristics of the ion beams used in these applications depend primarily on the source, which in this case is the plasma pinch. Therefore, it is necessary to study the specifications of the plasma pinch produced by two dense plasma focus devices with different operating energy and geometric when the nitrogen gas pressure was changed. The nitrogen gas was chosen due to its many uses, especially in the deposition of thin films, where it is used from the dense plasma focus as one of the methods for the manufacture of these films due to the good specifications shown by the thin films obtained this way.

2. Brief Description of the Methodology:

The Lee model code relates the electric circuit parameters with the thermodynamic, dynamic, and radiation of plasma focus. This code has been configured in 1983 and has been used for designing Mather-type plasma focus devices. The code has been improved to be five-phase by adding a small finite disturbance speed. The version including radiation and radiation-coupled dynamics was introduced and published in 2000. In 2007, plasma self-absorption was included. The code is widely used as a complementary facility in numerous. The main information obtained from the code

includes the axial and radial dynamics of the current sheath, SXR emission and its yield and total neutron yield. The code was used to determine current and neutron yield limitation, investigate neutron saturation effect, investigate radiative collapse and obtain anomalous resistance data. The model then has been also enhanced and used to study ion beams emitted from plasma focus devices [16-22]. Using the Lee model code, numerical experiments for two devices: APF [23,24] and PF400 [25,26] operated with nitrogen gas were simulated by entering several parameters including: Bank parameters (capacitor bank C_0 , inductance L_0 , stray circuit resistance r_0), Tube parameters (length z_0 and radius a of the anode, the radius of the cathode b) and Operational parameters (operator charging voltage V_0 and pressure P_0 of the gas, the energy storage in the capacitor bank E_0). The properties of the plasma pinch formed at the end of the focusing process were studied (pinch current, pinch length, pinch radius, pinch duration, and induced voltage within the pinch). These properties were chosen due to their effect on the properties of the ion beam emitted by the dense plasma focus device after the collapse of the pinch.

3. Numerical experiments and discussion

a. PF400 dense plasma focus device:

Numerical experiments have been done using the parameters shown in table 1 [27].

Table (1) PF400 dense plasma focus device parameters

Bank parameters	Static inductance $L_0 = 40 \text{ nH}$	Capacity $C_0 = 1 \text{ } \mu\text{F}$	stray resistance $r_0 = 10 \text{ m}\Omega$	
Tube parameters	cathode radius $b = 1.6 \text{ cm}$	anode radius $a = 0.6 \text{ cm}$	anode length $z_0 = 1.7 \text{ cm}$	
Operation parameters	Voltage $V_0 = 28 \text{ kV}$	Pressure $P_0 = \text{variable}$		
Model parameters	$f_m = 0.07$	$f_c = 0.7$	$f_{mr} = 0.13$	$f_{cr} = 0.7$

The computed current and voltage waveforms for PF400 device at 2 Torr of nitrogen are presented in the figure 2. The peak current is around 130 kA and the induced voltage is obtained to be 26 kV with the pinch duration of 4 ns. The pinch current, length and radius are 84 kA, 0.9 cm and 0.06 cm, respectively.

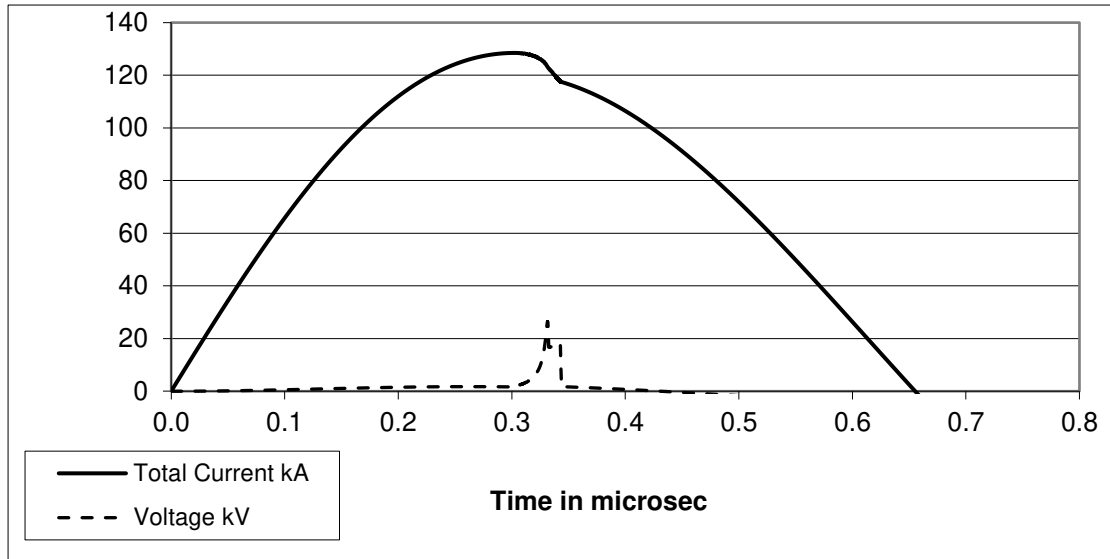


Figure (2) Total current and voltage waveforms for PF400 device at 2 Torr

After that many numerical experiments are carried out for different gas pressures. The pinch current, pinch length, pinch radius, pinch duration, and induced voltage versus pressure are studied and shown in the following figures.

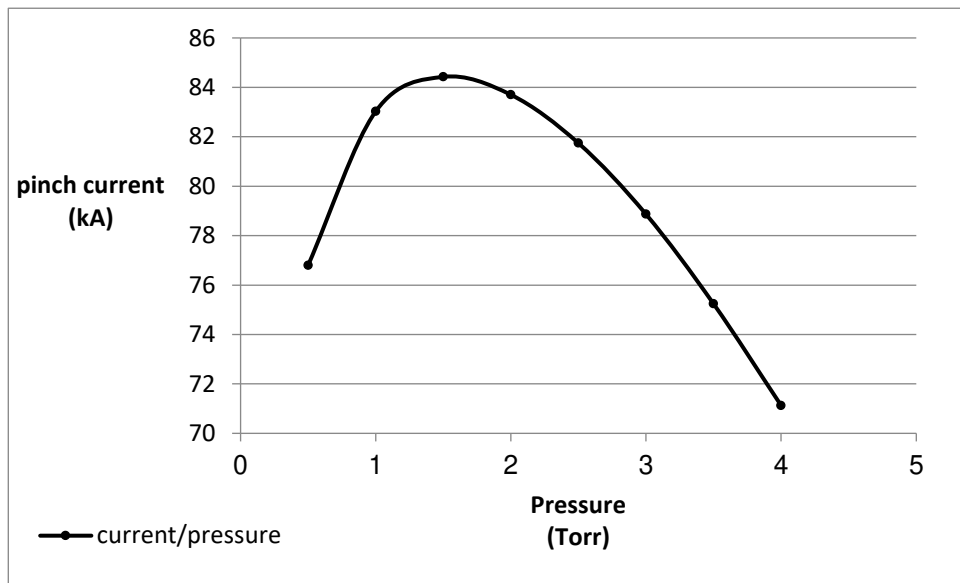


Figure (3) The changes of pinch current versus nitrogen pressure changes in PF400 device

Pinch current: We noticed that the pinch current increases with increasing pressure until it reaches a maximum value (84 kA) and then continues decreases as the pressure continues increasing due to the shifting of the focusing time towards the time of peak current.

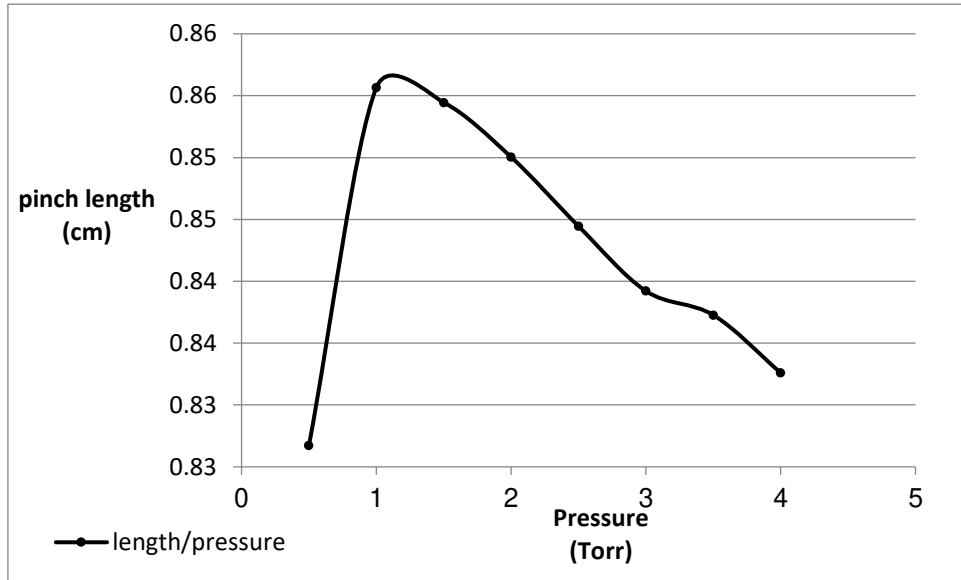


Figure (4) The change of pinch length versus nitrogen pressure changes in PF400 device

Pinch length: We noticed that the pinch length increased with increasing pressure until it reached a maximum value (0.85 cm) and then began to decrease as the pressure continued to increase.

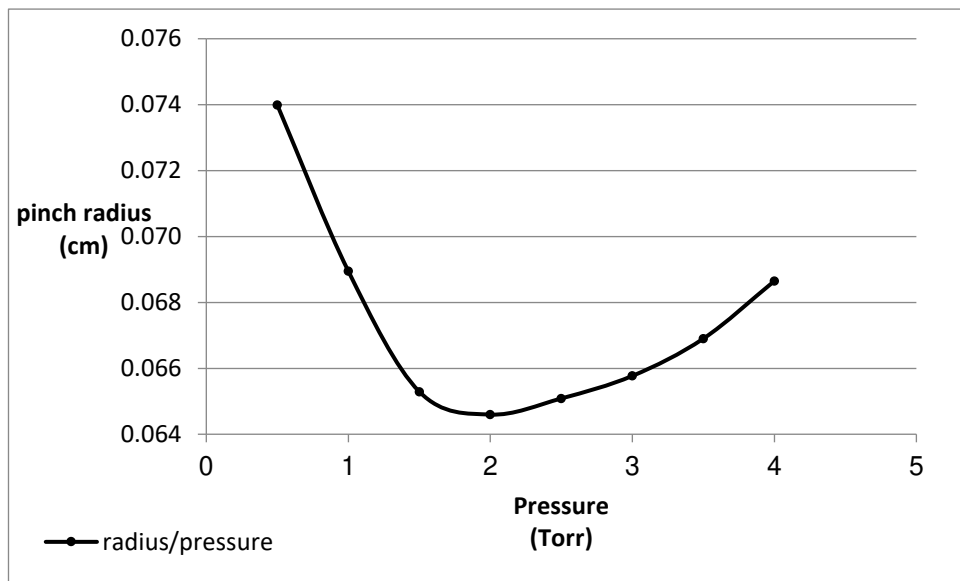
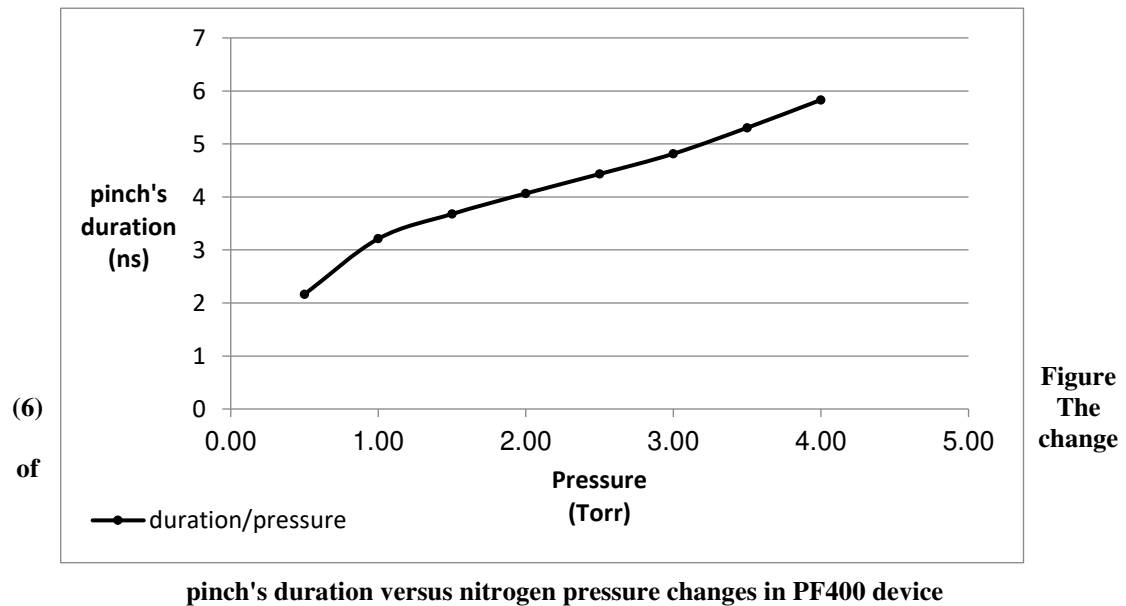


Figure (5) The change of pinch radius versus nitrogen pressure changes in PF400 device

Pinch radius: It is evident from Figure (5) that at the pressure value (1 Torr), the value of the pinch radius is (0.068 cm) and it starts decreasing and continues with increasing pressure until reaching the lowest value (0.064 cm) at (2 Torr) and then start increasing as the pressure continued to increase.

The explanation for these two opposite behaviors for the change in the length and radius of the plasma pinch with increasing pressure of nitrogen gas is to reach the compression phase at pressure (2 Torr), where the plasma pinch is in the phase of the strongest contraction and occurs in this phase most of the processes of radiation emission before the collapse of the pinch later, and this can be seen from the increase in the value of the radius of the pinch after this value of the pressure. That is, at the pressure value (1 Torr), the plasma pinch is in the vertical elongation phase, and with the continued increase in pressure, its length decreases and narrows.



Pinch duration: We notice that the pinch duration increases from the value (2.2 ns) at (0.5 Torr) and increases with the increase in nitrogen pressure until it reaches its highest value (5.8 ns) at (4 Torr).

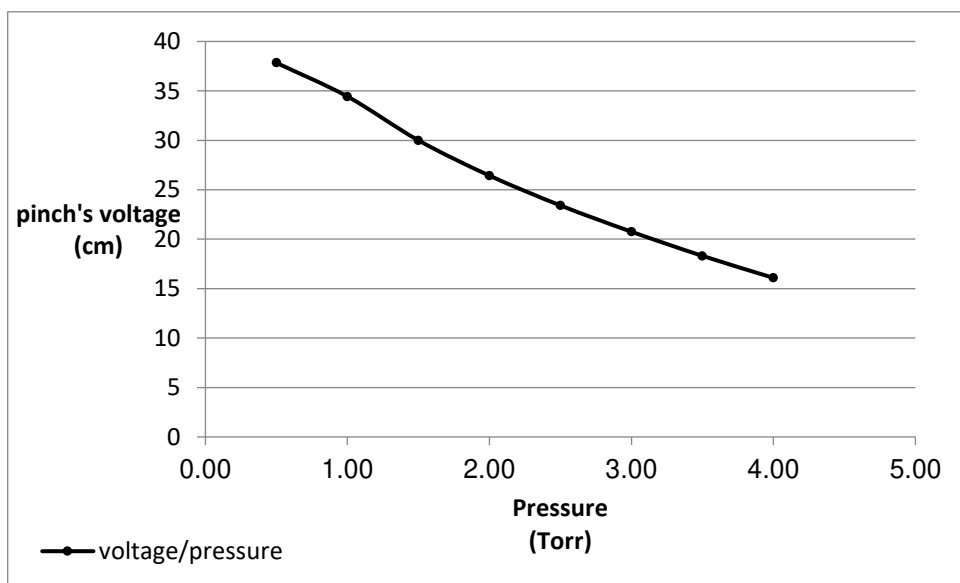


Figure (7) The change of induced voltage within plasma pinch versus nitrogen pressure changes in PF400 device

Induced voltage: Voltage starts from (38 kV) at (0.5 Torr) and decreases as pressure increases continue until (16 kV) at (4 Torr). This decrease in the value of the voltage inside the pinch can be explained by the decreasing in the energy coming from the capacitor bank to the pinch (thus the energy carried by the ions after the collapse of the pinch decreases as well).

b. APF dense plasma focus device:

Numerical experiments on APF device [24] have been done using the parameters shown in table 2.

Table (2) APF dense plasma focus device parameters

Bank parameters	Static inductance $L_0 = 145 \text{ nH}$	Capacity $C_0 = 36 \text{ }\mu\text{F}$	stray resistance $r_0 = 8 \text{ m}\Omega$	
Tube parameters	cathode radius $b = 2.235 \text{ cm}$	anode radius $a = 1 \text{ cm}$	anode length $z_0 = 14.8 \text{ cm}$	
Operation parameters	Voltage $V_0 = 12 \text{ kV}$	Pressure $P_0 = \text{variable}$		
Model parameters	$f_m = 0.02$	$f_c = 0.7$	$f_{mr} = 0.15$	$f_{cr} = 0.7$

By plotting the changes of each feature in terms of the applied gas pressure, we get the following figures:

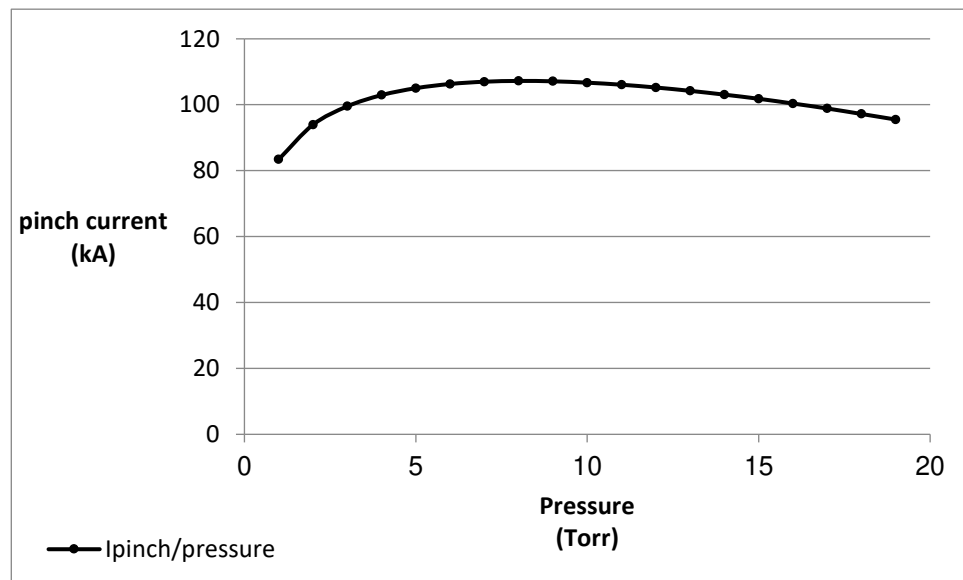


Figure (8) The change in the pinch current versus nitrogen pressure changes in APF device

Pinch current: We notice from Figure (8) an increase of pinch current in APF device from (83 kA) at (1 Torr) and increases with increasing pressure until reaching a maximum value (107 kA) at (7 Torr) and then decreases as the pressure continues to increase until (96 kA) at (19 Torr).

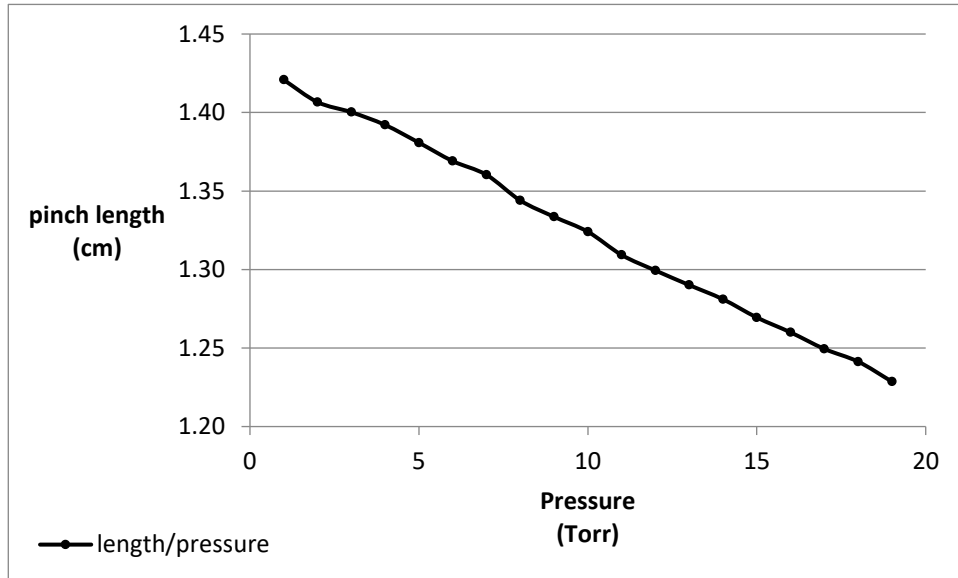


Figure (9) The change of pinch length versus nitrogen pressure changes in APF device

Pinch length: We notice from Figure (9) a decrease in the value of the plasma pinch length formed in the plasma focus device APF from (1.42 cm) at pressure (1 Torr) to (1.23 cm) at (19 Torr).

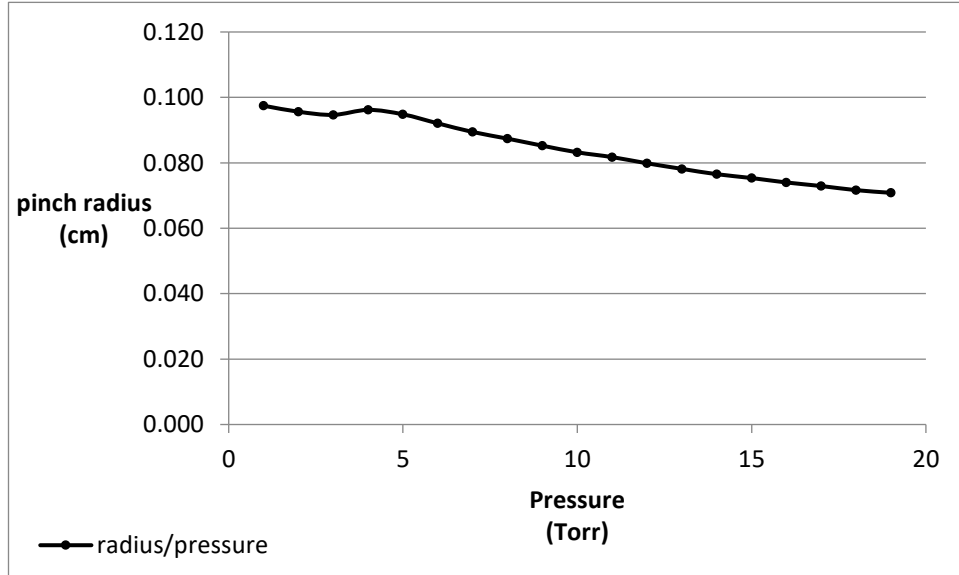


Figure (10) The change of pinch radius versus nitrogen pressure changes in the APF device

Pinch radius: pinch radius starts from (0.097 cm) at pressure (1 Torr) and decreases as pressure continues to increase until (0.071 cm) at pressure (19 Torr).

We notice from Figure (10) that there is a distinct value for the radius of the pinch at pressure (3 Torr), where there is a decrease in the value of the radius of the pinch as a

narrowing of the pinch occurs, and thus the pressure value at which most of the processes of radiation emission occur.

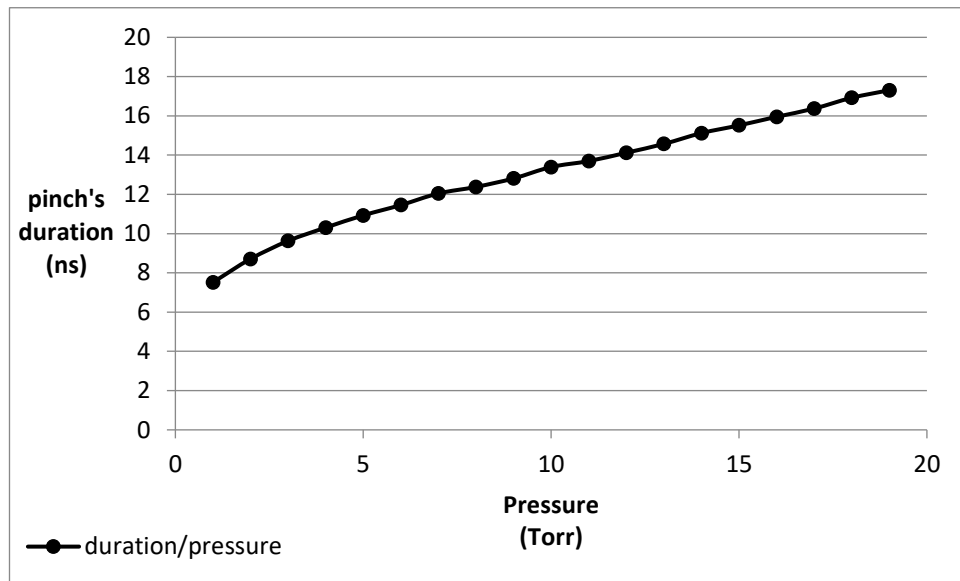


Figure (11) The change of pinch's duration versus nitrogen pressure changes in APF device

Pinch duration: pinch's duration starts from value (7.5 ns) at pressure (1 Torr) and increases with pressure up to value (17.3 ns) at pressure (19 Torr).

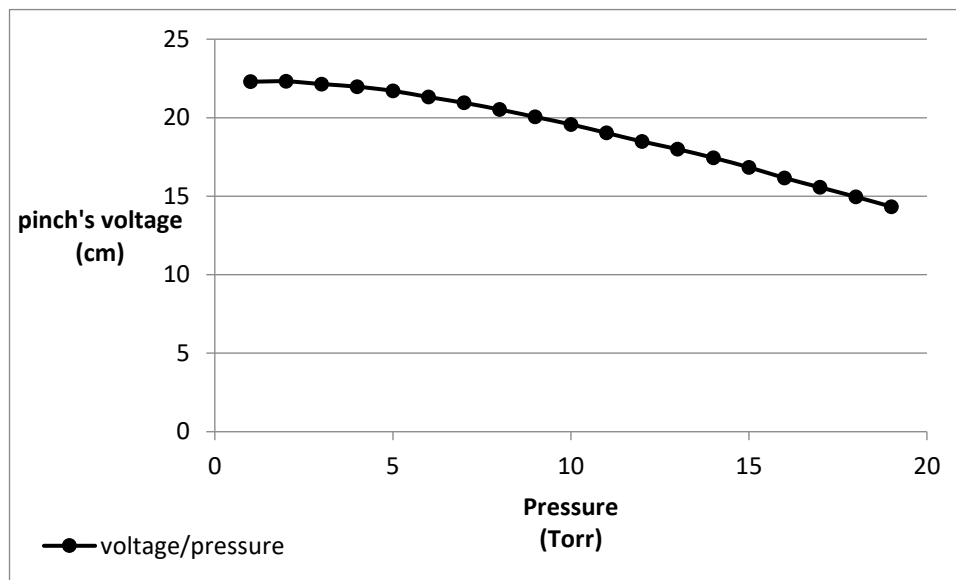


Figure (12) The change of Induced voltage versus nitrogen pressure changes in APF device

Induced voltage: The value Induced voltage in the pinch decreases from value (22 kV) to value (14 kV). This decrease in voltage is due to the decrease in the value of the power coming from the capacitor bank to the pinch phase.

c. Effect of the pinch properties on the number and energy of the ion beam:

The effect of the properties of the nitrogen plasma pinch (pinch length, radius, duration, voltage inside the pinch) on the number of nitrogen ions released by the pinch after its collapse and the energy carried by these ions was studied.

Number of ions beam [11] is calculated by multiplying the fluence of the ions by the cross-section of the plasma pinch:

$$N_i = 2.75 \times 10^{15} \frac{f_e}{\sqrt{M \cdot Z_{eff}}} \frac{\ln\left(\frac{b}{r_p}\right)}{r_p^2} \frac{I_{pinch}^2}{\sqrt{U}} \cdot \tau \cdot \pi r_p^2$$

where: f_e is the fraction of the plasma pinch energy carried by the ions, M is the mass of the ion, Z_{eff} is the effective charge, b is the radius of the cathode, r_p is the pinch radius, I_{pinch} is the pinch current, U is the acceleration voltage due to the collapse of the plasma pinch $U \approx 3V$ where V is the voltage generated within the plasma pinch, τ is the pinch duration.

Ions beam energy [11] is calculated by multiplying the number of ions beam by the effective charge and the voltage generated within the pinch according to the relationship:

$$E_{beam} = N_i \times Z_{eff} \cdot U$$

We note from these relations the dependence of the number of ions beam and their energy on the properties of the plasma pinch. So, using the Lee code, a series of numerical experiments were conducted on the two studied devices in order to find the effect of the properties of the nitrogen plasma pinch on the number and energy of the emitting ion beam after the collapse of the plasma pinch, according to the following:

The number of nitrogen ions beam: The change in the number of nitrogen ions emitted from the two studied devices was studied, changing the gas pressure, and the results were as shown in Figure (12):

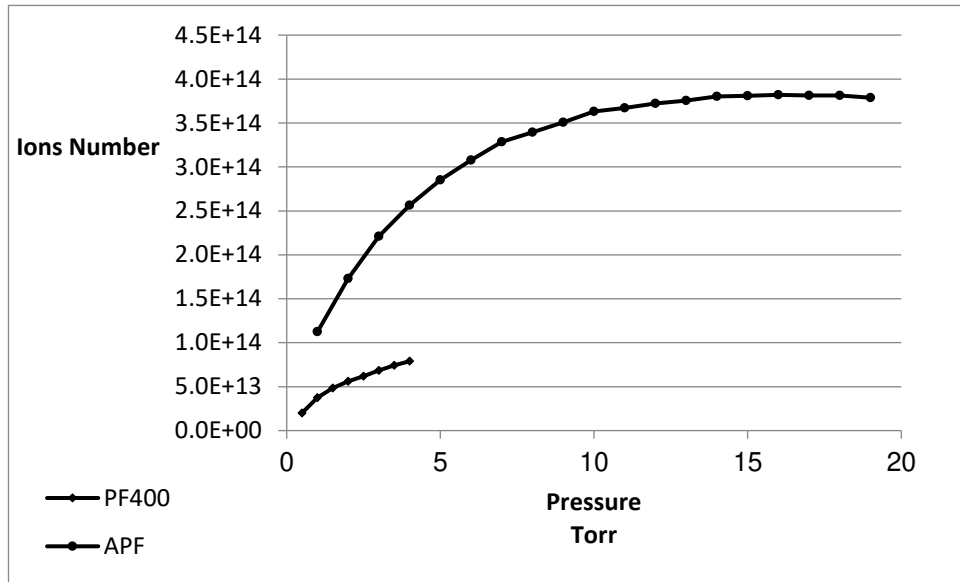


Figure (13) The change in the number of nitrogen ions emitted by both devices

In both devices, we notice an increase in the number of emitting ions with an increase in the pressure of the nitrogen gas, while noting the maximum number of ions in the APF device is 3.82×10^{14} ions while at PF 400 device is 7.9×10^{13} ions, due to the size of the formed plasma pinch and the value of the dimensions of the vacuum chamber, which led to an increase in the volume of ions confined within the pinch and as a result the high number of ions released after the collapse of the pinch.

Variation of the beam energy: By comparing the energy of the ion beam emitted by both devices, we obtained the following results shown in Figure (14):

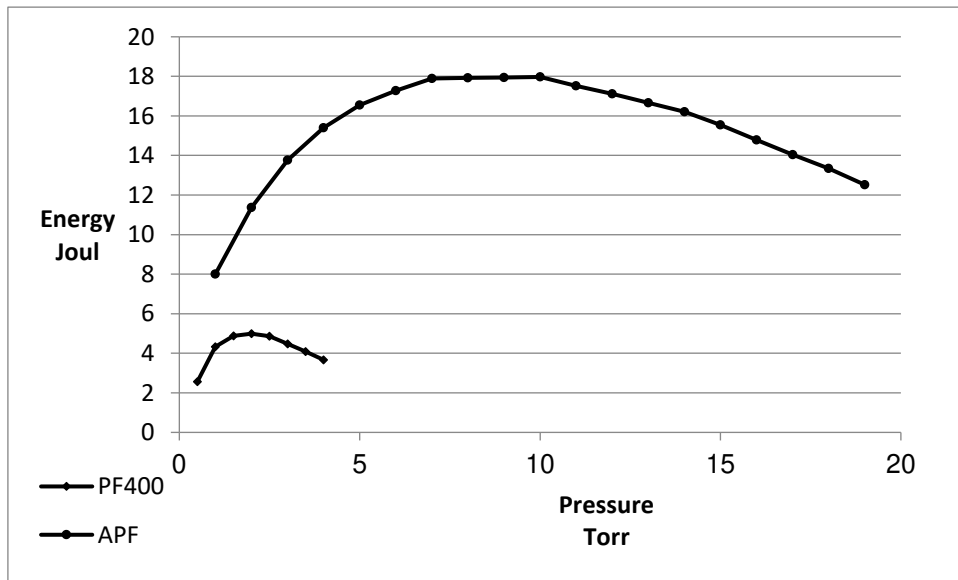


Figure (14) The energy change of the ion beam in both devices

We observe the same behavior in both devices where the beam energy increases with increasing pressure until reaching a maximum value, and we also note the value of the ion beam energy in the APF device (18 J) due to the large number of beam ions

compared to the PF 400 device (5 J) although the voltage generated within the pinch of the PF 400 device was larger.

The effect of changing the properties of the studied plasma pinch on the number of ions released after the collapse of the pinch was studied, as the change in the number of ions with the change in gas pressure had the same behavior, noting the high number of ions in the APF device compared to the PF 400 device, and that Due to the influence of several factors, such as an increase in the size of the pinch (length and radius) and consequently the number of ions trapped within it, as well as an increase in the retention period of the pinch and the value of the high current of the pinch, in addition to a decrease in the value of the latency generated within the pinch, where the number of issued ions is inversely proportional to the value of the voltage generated within pinch.

It should be noted that the Lee code does not include all phenomena that occur within the plasma focus discharge, e. g. plasma/current disruptions within the plasma column, center-peaking density effects, localized regions of high density and temperature. But an average density and temperature (uniform in the entire pinch volume), which are probably well below the measured peak density and temperature, are computed. However, since the 4 model parameters (f_m , f_c , f_{mr} , f_{cr}) are obtained, the model incorporates approximately energy and mass balances equivalent for all processes that are not even specifically modeled within the code.

4- conclusion:

This study provided explanations for the change in the properties of plasma pinch when changing the pressure of nitrogen gas in two dense plasma focus devices with different operating energy and geometric dimensions and the effect of these changes on the number and energy of ions emitted after the collapse of the pinch, and thus can be used in various applications such as the synthesis of thin films. The study can be expanded later to include other plasma focus devices when using different gases to suit the required practical application.

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Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Data availability:

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Author Contributions:

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by [Alaa Nassif], [Walid Sahyouni] and [Ola Zeidan]. The first draft of the manuscript was written by [Alaa Nassif] and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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